Project

🎤 ✅ 1. PERFECT 2–3 MIN PROJECT EXPLANATION

> "In my current role, I worked on a commingle data synchronization project for Elevance Health, where we ensured Deductible and Out-of-Pocket accumulations were consistent between internal systems and external vendors like pharmacy, dental, and mental health providers.
>
> We built an end-to-end pipeline using AWS and Snowflake with a Medallion architecture.
>
> For the inbound process, vendors send CSV files which are stored in Amazon S3. We used IAM role-based storage integration to securely connect Snowflake with S3. Snowpipe with auto-ingest loads the data into the Bronze layer in near real-time.
>
> Then we implemented CDC using Snowflake Streams to capture incremental changes and Tasks to process those changes into the Silver layer, where we handle deduplication, validation, and transformations using SQL.
>
> In the Gold layer, we aggregate data to calculate member-level Deductible and Out-of-Pocket values.
>
> For the outbound process, we export the processed data back to S3 using COPY INTO, and vendors consume this data to keep their systems synchronized.
>
> > To ensure reliability, we implemented monitoring and alerting mechanisms. We tracked Snowpipe loads using COPY_HISTORY and Task executions using TASK_HISTORY. Based on success or failure conditions, alerts were triggered via AWS SNS to notify the team through email/Slack, enabling quick issue resolution and proactive monitoring.

🧱 ✅ 2. ARCHITECTURE (EXPLAIN STEP BY STEP)

🔄 Flow

Vendor → S3 → IAM → Snowpipe → Bronze → Stream → Task → Silver → Gold → S3 → Vendor

🔹 Step 1: Data Source

- Vendors send **CSV files**
- Stored in:

s3://elevance-inbound/vendor_data/

🔹 Step 2: Secure Access (IAM)

> "We used IAM roles for secure access."

```
CREATE STORAGE INTEGRATION s3_int

TYPE = EXTERNAL_STAGE

STORAGE_PROVIDER = S3

ENABLED = TRUE

STORAGE_AWS_ROLE_ARN =
'arn:aws:iam::xxx:role/snowflake-role'

STORAGE_ALLOWED_LOCATIONS =
('s3://elevance-inbound/');
```

🔹 Step 3: Snowflake Stage

```
CREATE STAGE vendor_stage

URL = 's3://elevance-inbound/vendor_data/'

STORAGE_INTEGRATION = s3_int;
```

🔹 Step 4: Snowpipe (Real-time Ingestion)

```
CREATE PIPE vendor_pipe

AUTO_INGEST = TRUE

AS

COPY INTO bronze.vendor_raw

FROM @vendor_stage

FILE_FORMAT = (TYPE = 'CSV');
```

🔹 Step 5: Bronze Layer

> "Stores raw data as-is"

- No transformation
- Used for audit

🔹 Step 6: CDC (Streams)

```
CREATE STREAM vendor_stream ON TABLE
bronze.vendor_raw;
```

> "Tracks only incremental changes"

🔹 Step 7: Processing (Task + MERGE)

```
MERGE INTO silver.vendor_clean t

USING (

  SELECT *, ROW_NUMBER() OVER
(PARTITION BY member_id ORDER BY
update_ts DESC) rn

  FROM vendor_stream

) s

ON t.member_id = s.member_id

WHEN MATCHED AND s.rn = 1 THEN

  UPDATE SET t.ded = s.ded, t.oop = s.oop

WHEN NOT MATCHED AND s.rn = 1 THEN

  INSERT VALUES (s.member_id, s.ded,
s.oop);
```

🔹 Step 8: Silver Layer

> Cleaned + deduplicated + validated data

🔹 Step 9: Gold Layer

```
CREATE TABLE gold.member_accum AS

SELECT member_id,

       SUM(ded) AS total_ded,

       SUM(oop) AS total_oop

FROM silver.vendor_clean

GROUP BY member_id;
```

🔹 Step 10: Outbound

```
COPY INTO @vendor_stage/outbound/

FROM gold.member_accum

FILE_FORMAT = (TYPE = 'CSV')

SINGLE = TRUE;
```

🔔 ✅ 3. ALERTING (ADD THIS TO SOUND STRONG)

> "We implemented alerts using AWS SNS and Snowflake task monitoring."

- Success & failure alerts
- COPY history monitoring
- Task history tracking

⚡ ✅ 4. PERFORMANCE OPTIMIZATION

Say this:

- "We used clustering on member_id"
- "Used Streams instead of full loads"
- "Optimized SQL queries"
- "Used appropriate warehouse sizing"

🧠 ✅ 5. COMMON FOLLOW-UP QUESTIONS

❓ Why Snowpipe?

👉 "For near real-time ingestion without managing compute"

❓ Why Streams?

👉 "To process only changed data (CDC)"

❓ Why Task?

👉 "To automate incremental processing"

❓ How did you handle duplicates?

👉 "Using ROW_NUMBER and keeping latest records"

❓ How do you debug failures?

👉

- COPY_HISTORY
- VALIDATION_MODE
- Task history

❓ Why Medallion architecture?

👉

> "To separate raw, cleaned, and business layers for better data quality and maintainability"

🎯 ✅ 6. SHORT VERSION (30 SECONDS)

> "We built an AWS + Snowflake pipeline where vendor CSV files are ingested into S3, loaded into Snowflake using Snowpipe, processed incrementally using Streams and Tasks, transformed into Silver and Gold layers, and exported back to S3 for vendor synchronization."

🔥 FINAL TIP

👉 Always say:

- "In our project…"
- "We implemented…"
- "We optimized…"

This shows **real experience**

🚀 NEXT STEP

If you want, I can:

- 🎤 Do **mock interview on this project**
- 💡 Add **real production challenges**
- 📄 Convert into **resume project bullet points**

👉 Just say:

"Take mock interview" 🔥